

SPATIOTEMPORAL VARIABILITY OF SNOW DEPLETION CURVES DERIVED FROM SNODAS FOR THE CONTERMINOUS UNITED STATES, 2004-2013¹

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ABSTRACT: Assessment of water resources at a national scale is critical for understanding their vulnerability to future change in policy and climate. Representation of the spatiotemporal variability in snowmelt processes in continental-scale hydrologic models is critical for assessment of water resource response to continued climate change. Continental-extent hydrologic models such as the U.S. Geological Survey National Hydrologic Model (NHM) represent snowmelt processes through the application of snow depletion curves (SDCs). SDCs relate normalized snow water equivalent (SWE) to normalized snow covered area (SCA) over a snowmelt season for a given modeling unit. SDCs were derived using output from the operational Snow Data Assimilation System (SNODAS) snow model as daily 1-km gridded SWE over the conterminous United States. Daily SNODAS output were aggregated to a predefined watershed-scale geospatial fabric and used to also calculate SCA from October 1, 2004 to September 30, 2013. The spatiotemporal variability in SNODAS output at the watershed scale was evaluated through the spatial distribution of the median and standard deviation for the time period. Representative SDCs for each watershed-scale modeling unit over the conterminous United States ($n = 54,104$) were selected using a consistent methodology and used to create categories of snowmelt based on SDC shape. The relation of SDC categories to the topographic and climatic variables allow for national-scale categorization of snowmelt processes.

(KEY TERMS: snow hydrology; geospatial analysis; watersheds; remote sensing.)

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INTRODUCTION

National-scale assessment of water resources, including water availability, is critical for understanding the vulnerability of these resources to changes in policy and climate. Water resources assessment for the conterminous United States (CONUS) requires the availability of hydrologic information derived using a consistent, standardized methodology. Small, watershed-scale study area projects provide critical

information toward understanding the complexity of hydrologic processes; however, the ability to translate, generalize, and compare hydrologic processes at a continental-scale is a critical step in the field of hydrology (McDonnell *et al.*, 2007; Archfield *et al.*, 2015). This study focuses on improving the representation of snowmelt processes at the CONUS scale.

Snow processes are often studied and modeled at the watershed- or field-scale due to their complexity and local factors (such as wind redistribution and vegetation) which influence snowpack accumulation

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and snowmelt (Molotch, 2009; Homan *et al.*, 2011; Jepsen *et al.*, 2012; Broxton *et al.*, 2014). Computationally-expensive models which account for these factors are useful at watershed or field scales; however, estimation of snow processes on a continental scale requires simplification of these complex processes. Snow depletion curves (SDCs) are a simple and dynamic method used in hydrologic models to describe snowmelt processes (Martinec and Rango, 1981; Liston, 1999; Markstrom *et al.*, 2015). SDCs relate snow covered area (SCA) to snow water equivalent (SWE) for a given hydrologic response unit (HRU) over a snowmelt season (Luce and Tarboton, 2004; Kolberg and Gottschalk, 2006; Fassnacht *et al.*, 2015). Snowmelt is a unidirectional process over time, so SDCs inherently include a temporal factor. To create comparative SWE and SCA relations, these data can be normalized to percent of total area and percent of peak SWE, respectively. Conceptually, the shape of an SDC integrates the spatiotemporally-complex process of snowmelt, using SWE and SCA (Figure 1). SDCs allow for a comparable relation over space and time of snowmelt processes.

Investigation of snowmelt processes in the U.S. and their connection to climate change are concentrated in the Western U.S. (Cline *et al.*, 1998; Clow, 2010; Harpold *et al.*, 2014; Dettinger *et al.*, 2015; Barnhart *et al.*, 2016; Broxton *et al.*, 2016), where water resources are snowmelt-dominated (Doesken and Judson, 1996). Data sources for snowmelt are often fixed at station locations, such as the U.S. Department of Agriculture National Resources Conservation Service's Snow Telemetry (SNOTEL) locations. Methodologies for spatial interpolation of accumulated snowfall and SWE from these station data to larger areas have been analyzed, using statistical methods and evaluated using the CONUS-scale Snow Data Assimilation System (SNODAS) as a comparison (Fassnacht *et al.*, 2003; Broxton *et al.*, 2016). SNOTEL stations are concentrated in the Western U.S., neglecting snow

processes in the Eastern U.S., which are not insignificant (Markstrom *et al.*, 2016) and critical to a national-scale assessment of water availability.

The objective of this study is to develop and assess SDCs derived from SNODAS output, using a nationally consistent methodology. Gridded daily SWE and calculated SCA were spatially aggregated to 109,951 watershed-scale HRUs over CONUS for each snowmelt season within the study period (2004–2013). The inter-annual variability in SNODAS output and resulting SDCs were assessed for each HRU. For each HRU, a representative SDC was selected where appropriate and used to compare snowmelt processes over the CONUS. The development of SDCs by HRU will improve current continental-scale representation of snow processes, which are currently very limited (*e.g.*, two SDCs, one for HRUs above treeline and one for below) (Markstrom *et al.*, 2015). Categorization and relation of SDCs to geographic variables will allow for translation of snowmelt process variability to data-limited snowmelt-dominated areas (such as Alaska).

STUDY AREA AND DATA SOURCES

Study Area and Geospatial Fabric

The spatial extent of this study extends over the entirety of the CONUS. In addition to gridded SWE and SCA, geographic variables (elevation and slope) were aggregated to the 109,951 HRUs of the Geospatial Fabric for the National Hydrologic Model (NHM) (Viger and Leavesley, 2007; Viger, 2014; Viger and Bock, 2014) (Figures 2A and 2B). While these variables were designed specifically to run the U.S. Geological Survey (USGS) Precipitation-Runoff Modeling System (PRMS) (Markstrom *et al.*, 2015), the methods to spatially-aggregate these temporally-static

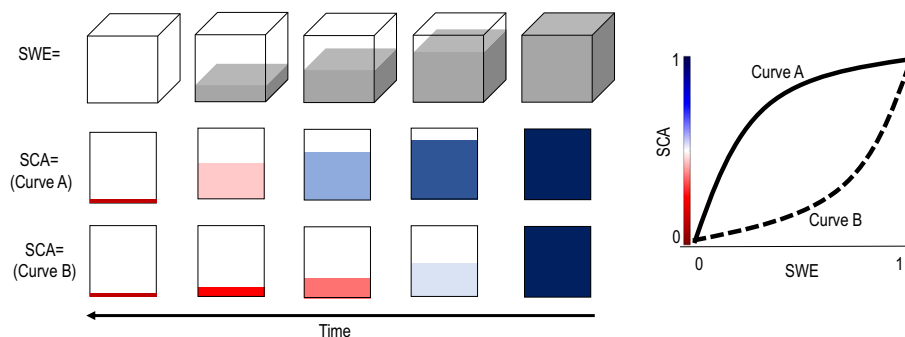


FIGURE 1. Conceptual Model of Snapshots in Time of Normalized Snow Water Equivalent (SWE) and Snow Covered Area (SCA) for Two Different Snowmelt Systems (Curve A and Curve B) at Left, and These Curves Shown as Snow Depletion Curves (SDCs) at Right. Colors represent SCA values, from 0 to 1, as shown on the SDC plot (right). The progression of time over the snowmelt season is from 1 (maximum normalized SWE and SCA) to 0 (left to right).

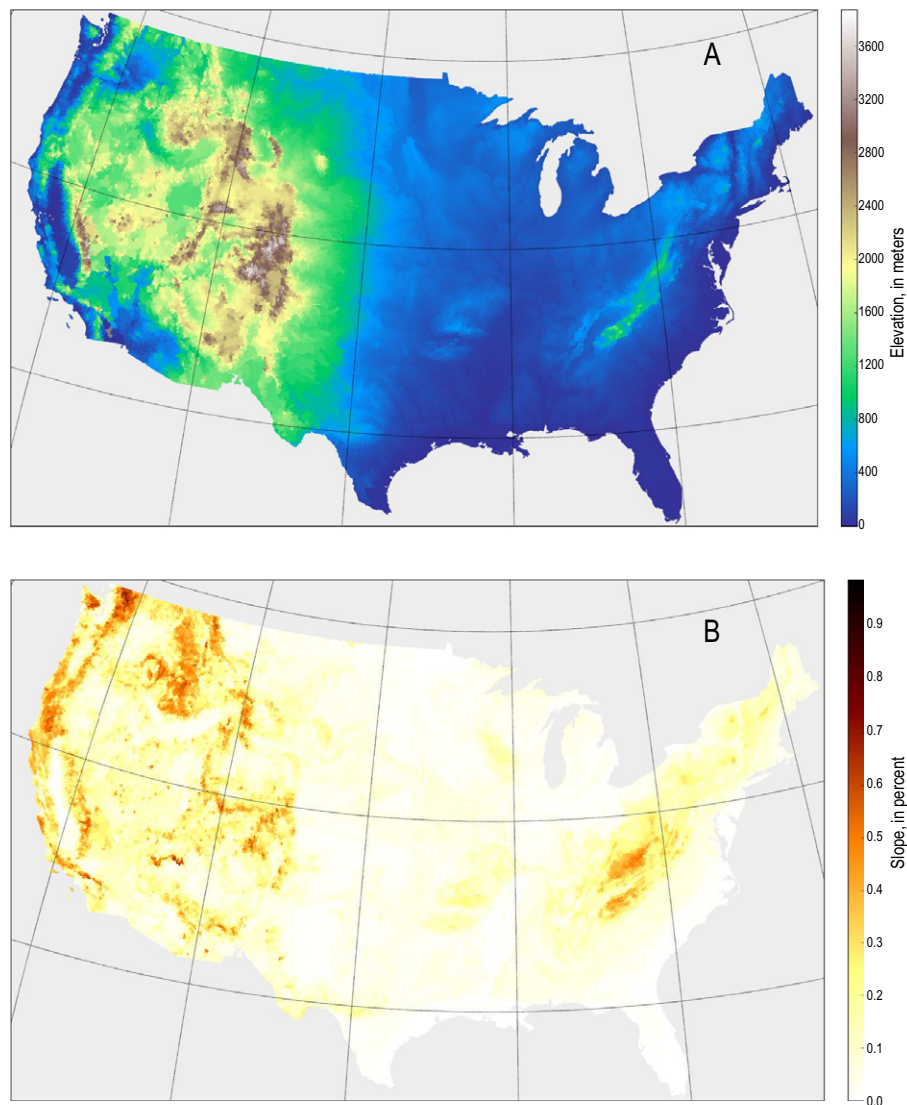


FIGURE 2. Spatial Distribution of Elevation (A) and Slope (B) at the Hydrologic Response Unit (HRU)-Scale Over Conterminous United States (CONUS).

variables are directly translatable for use in this study. The HRUs are derived from the National Hydrography Dataset, NHDPlusV1 (U.S. Environmental Protection Agency and U.S. Geological Survey, 2005). Sizes of the HRUs are based on drainage density and range from 1-square kilometer (km^2) to over 67,000 km^2 , with an average size of 74 km^2 . There are contributing contiguous HRUs which span international borders, and these areas were not included in this study because they did not coincide with the spatial extent of SNODAS output.

SNODAS Output

This study used publically available, spatially distributed SNODAS data products created by the

National Operational Hydrologic Remote Sensing Center (NOHRSC), which are available through the National Snow and Ice Data Center (NSIDC). The NSIDC is jointly operated by the NOHRSC within the National Oceanic and Atmospheric Administration (NOAA) and National Aeronautics and Space Administration (NASA) (National Operational Hydrologic Remote Sensing Center, 2004). SNODAS output can be obtained online (<https://nsidc.org/data/g02158>).

SNODAS output integrates a combination of data sources (station, airborne, and satellite) in conjunction with downscaled output from the Numerical Weather Prediction models to produce SWE for the CONUS at a 1-km grid spatial and at a daily temporal resolution.

While field-scale, watershed, and regional studies have found discrepancies between SNODAS output and observed values (Clow *et al.*, 2012; Hedrick *et al.*,

2015), SNODAS output is the only currently available product at the CONUS extent. Other CONUS or global extent data products, such as NASA's Moderate Resolution Imaging Spectroradiometer (MODIS; <https://modis.gsfc.nasa.gov/>) provide SCA, but do not model or measure SWE. Development of SDCs, using SNODAS SWE and calculated SCA from the same SNODAS output allows for development of SDCs on a continental scale.

METHODS

SWE and SCA

Daily 1-km gridded SNODAS output from October 1, 2004 to September 30, 2013 (the study period) provided the SWE values used for this study. The spatial and temporal components of the SNODAS output and methodologies used to calculate SCA from SWE are described further in the following subsections.

Spatial Assessment. Spatial aggregation of gridded SNODAS output to the HRU scale was completed, using the USGS Geo Data Portal (GDP),

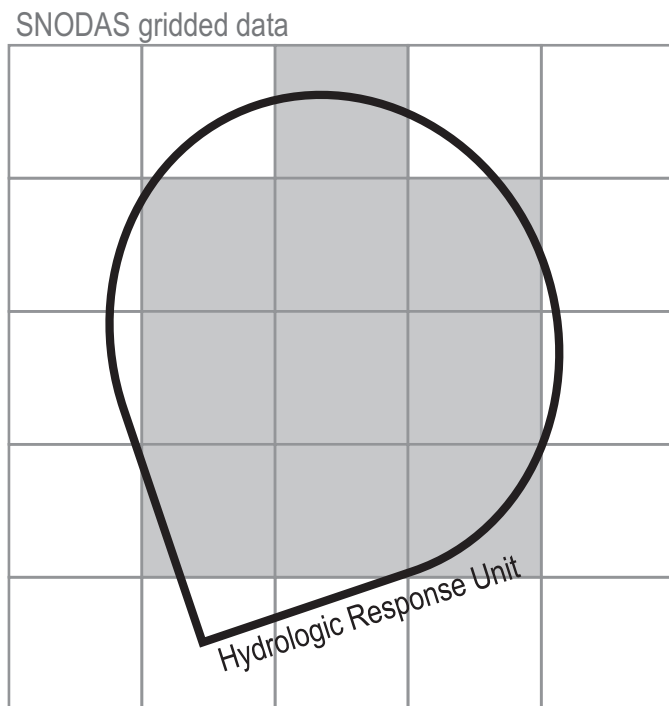


FIGURE 3. Diagram Showing How Gridded Snow Data Assimilation System (SNODAS) Output Were Aggregated to the Hydrologic Response Unit Scale.

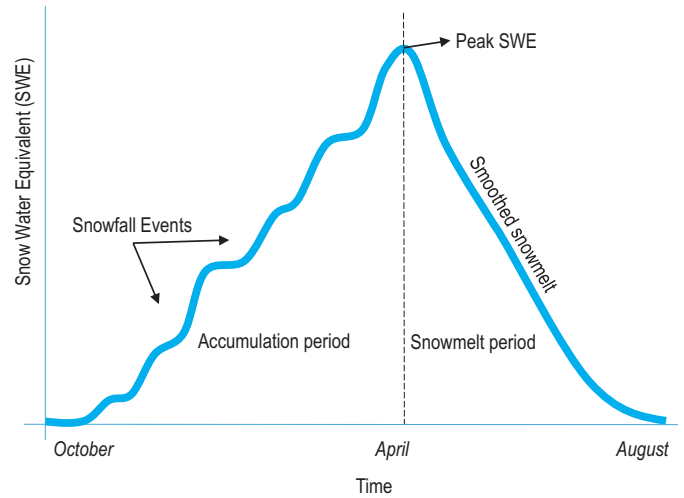


FIGURE 4. Idealized Accumulation and Melt of a Snowpack Over Time after Reversals Have Been Removed.

<http://cida.usgs.gov/gdp/> (Blodgett *et al.*, 2011). The GDP summarized the daily time series of SNODAS SWE for each HRU in the CONUS. Because most HRUs cover multiple SNODAS grid cells (Figure 3), the GDP calculated an area-weighted mean of the daily gridded SWE values within each HRU. SCA is not included as one of the base snow parameters of the SNODAS output, so an estimate of SCA for each grid cell was calculated using a binary cell assignment: 1 if SWE > 0 and 0 if SWE = 0. This calculated gridded SCA was then aggregated to the HRU scale by calculating the number of cells equal to 1 divided by total number of cells covered by each HRU for each day.

Temporal Assessment. Daily time series of SWE and SCA over the study period were used to develop SDCs for each HRU within the CONUS (Figure 4). The timing and duration of the snowmelt season were calculated for each HRU and for each season as the date of peak SWE and the number of days between that date and the day on which SWE = 0. The standard deviation of the annual peak SWE dates and snowmelt season durations were also calculated for each HRU for the study period.

Description of Variables

SNODAS-derived variables were derived from SWE and calculated SCA values at the HRU-scale. Digital elevation model (DEM)-derived variables were derived as part of the NHM Geospatial Fabric (Viger, 2014). SNODAS- and topographically derived variables, their data sources, definitions, and units are summarized in Table 1.

TABLE 1. Variables Used in Analysis of Snow Depletion Curves.

Variable Name	Data Source	Definition	Units
Elevation	DEM	The median elevation value within each HRU. See Viger, 2014 for method details	Meters
Slope	DEM	The area-averaged slope for each HRU. See Viger, 2014 for method details	Percent
Median SWE	SNODAS	The median value of annual peak SWE values over the study period	Millimeters
Reversals	SNODAS	The cumulative number of snowmelt season snowfall events (SCA increases) for each HRU, cumulative for the study period	NA
Snowmelt timing	SNODAS	The median day of annual peak SWE for each HRU over the study period	Days since October 1
Snowmelt duration	SNODAS	The median of the number of days between snowmelt onset until SCA equals 0 for each HRU for each year of the study period	Days
Standard deviation of snowmelt timing	SNODAS	The standard deviation of annual day of peak SWE for each HRU over the study period	Days
Standard deviation of snowmelt duration	SNODAS	The standard deviation of the annual snowmelt season duration for each HRU over the study period	Days
Similarity	SNODAS	The sum of the vertical distances between annual SCA values and the median for each SWE value, divided by the number of SCA values for each HRU	Dimensionless

Note: SNODAS, Snow Data Assimilation System; SWE, snow water equivalent; SCA, snow covered area; HRU, hydrologic response unit.

Snow Depletion Curves

HRU-normalized (from 0 to 1) SCA and SWE data were plotted to generate annual SDCs for each year of the study period (Figure 1). This integrated measure of snowmelt is simplified to show two different snowmelt processes: Curve A represents a snowmelt process which loses SCA at a slower rate than SWE, and Curve B represents a snowmelt process which loses SCA at a faster rate than SWE. We expect that SDCs which are closer to Curve A to be in areas which have a consistent, homogeneous snowpack over the HRU. We expect SDCs closer to Curve B to be in areas with more spatially heterogeneous snowpack over the HRU.

Increases in SWE and SCA due to snowfall after peak SWE (during the snowmelt season) were counted (see “reversals” in Table 1) and removed from each curve. When SCA returned to the value prior to the increase, data were again included in the series. This method created SDCs which represent an idealized, uninterrupted melt, which allows for comparison of this process over space and time. While this method was useful for the purposes of this study, it also extended the length of the snowmelt season.

Selection of HRUs. An inclusive analysis over all of the CONUS to generate representative SDCs included a significant number of HRUs which should not be included in an analysis of snowmelt processes, such as HRUs which: do not receive snow; do not build a snowpack; and have highly variable SDCs from year to year. Representative SDCs were assigned to a subset of HRUs, using a series of six selection criteria. These criteria were used to remove HRUs with inadequate snowpack for representation with an SDC in an orderly, methodical way. The first criteria was applied to remove

all HRUs without full SNODAS coverage (those spanning international borders), leaving 109,544. The second criteria were applied to remove all HRUs without snow (no SWE), leaving 86,520. The third criteria were applied to remove small HRUs which had too few SNODAS grid cells (less than a 25 grid cell threshold), leaving 68,966. The fourth criteria was applied to remove any HRUs with areas dominated by water bodies (greater than a 50% threshold), leaving 68,584. The fifth criteria was applied to remove HRUs where only a portion of the HRU received snow (SCA values less than 50%), leaving 65,048. The sixth and last criteria was applied to remove any HRUs which are only partially covered with snow for the snowmelt season (in other words, where SCA values remain constant for over 80% of SWE values), leaving 56,374. This last criteria removes HRUs which span areas and (or) elevations where snow falls on only part of the HRU. The following analyses use this subset of HRUs ($n = 56,374$).

Selection of Representative SDC. A consistent method was developed to select a representative SDC from available annual SDCs for each HRU. The first step in this process was to develop a metric to determine the “similarity” of the SDCs of each snowmelt season over the study period (Equation 1):

$$\text{similarity} = \sum_0^n \frac{d_n}{p} \quad (1)$$

where n is the number of snowmelt seasons with SDCs, d is the distance in the y -axis (SCA) direction from the annual curve to the median curve along the SCA axis, and p is the number of points on the SDC curve. The similarity value measures the temporal variability in SDCs for each HRU. A zero value of

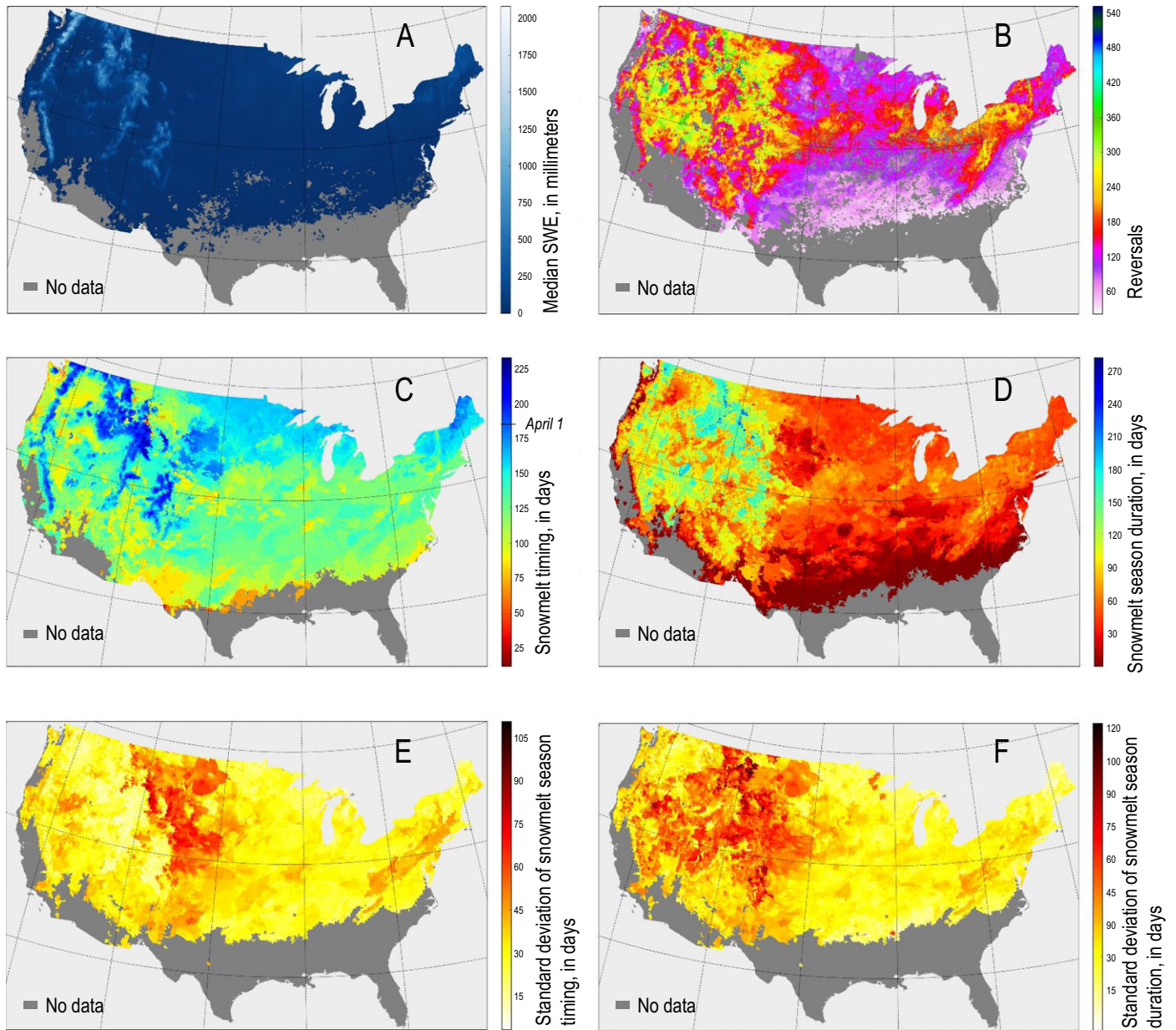


FIGURE 5. Spatial Distribution of HRU-Scale SNODAS-Derived Variables for the Study Period Over CONUS.

similarity represents an identical SDC, whereas higher values of similarity represent more dissimilar SDCs. HRUs with similarity values greater than a given threshold (0.15) were determined not to have similar enough curves to adequately be able to describe with a representative SDC, and were not included in further analysis. This threshold value was chosen to eliminate the most dissimilar SDCs ($n = 2,270$) for the current analysis; however, this threshold could be modified to constrain SDCs to be only the most similar. A single, representative SDC for each HRU was then selected from the nine SDCs. The representative SDC selected was the closest to the median SDC from each of the nine snowmelt

seasons in the study period, evaluated independently for each HRU. Representative HRU-scale SDCs were then compared and aggregated to assess spatial variability in SDCs across the CONUS ($n = 54,104$).

RESULTS

Spatial Distribution of HRU-Scale SNODAS-Derived Variables

The spatial distribution of six SNODAS-derived variables from Table 1 (median SWE, reversals,

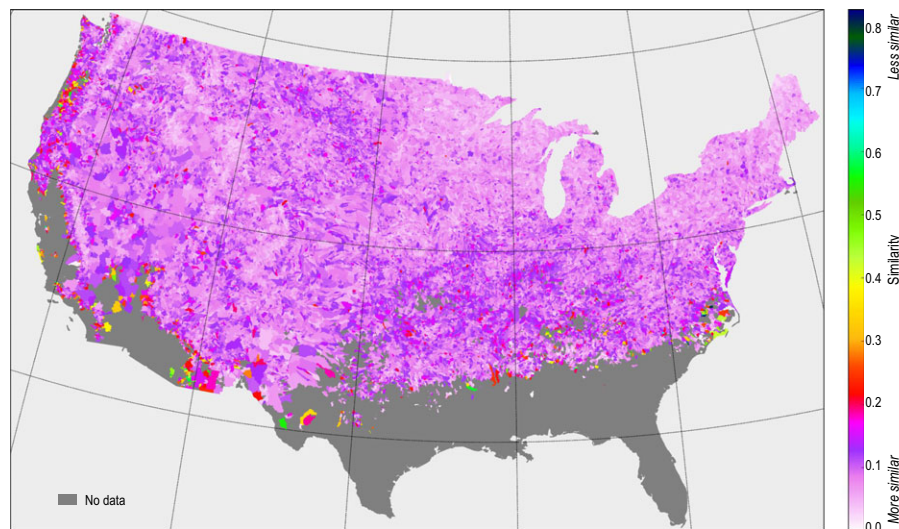


FIGURE 6. Spatial Distribution of Similarity of SDCs for the Time Period Over CONUS.

snowmelt timing, snowmelt duration, standard deviation of snowmelt timing, and standard deviation of snowmelt duration), show the spatial distribution of temporal variability in snowmelt processes at the CONUS scale (Figure 5). Higher median SWE values are located at higher elevation HRUs in the West, and higher latitude HRUs in the Midwest and Northeast U.S. (Figure 5A). Reversals show areas where there is more late-season snowfall, generally at higher elevation HRUs in the West, and also areas where there is more snowmelt season duration variability (Figure 5B). The median day of peak SWE shows later snowmelt at higher elevations in the West, and high latitude in the Midwest and Northeast U.S. (Figure 5C). The median snowmelt season duration shows higher values in the continental snowpacks relative to the coastal snowpacks in the West (Figure 5D). The standard deviation of the date of peak SWE shows greater variability in the interior, continental snowpacks in the West as well as the Appalachian Mountains in the East (Figure 5E). The standard deviation of the snowmelt season duration shows greatest variability in the snowpacks of the Sierra Nevada as well as the interior western snowpacks (Figure 5F). The total number of snowmelt reversals (increases in SWE after peak SWE occurs) is greatest where there is also high variability in snowmelt season duration in addition to some areas in the Midwest and the Appalachian Mountains (Figures 5B and 5F). The Great Plains region east of the Rockies has a high standard deviation of both day of peak SWE and number of days in the snowmelt season (Figures 5E and 5F).

Spatial Distribution of SDC Similarity

After SDCs were generated for each selected HRU for each snowmelt season in the study period a representative SDC was chosen. The temporal variability in the SDCs was assessed through a calculated “similarity” value, as previously described, the spatial distribution of which is shown in Figure 6. Low values of similarity show less inter-annual variability and high values show higher inter-annual variability. The most dissimilar areas are in the fringes of where SDCs were calculated, such as the South and coastal areas which do not consistently receive enough snow to build a snowpack. Similarity shows that there is less inter-annual variability in SDCs in the Northeast and northern Midwest, as well as the high-elevation areas where there is larger snowpack (Figure 6).

Spatial Distribution of SDCs

The representative SDC for each HRU was used to generate clustered SDCs to assess the spatial distribution of similar SDCs over CONUS (Figure 7). The relation of topographic and climatic variables was examined for HRUs with SDCs. SDCs were classified into three types: low SCA, mid-SCA, and high SCA. The classification was determined by the value of SCA at 50% normalized maximum SWE (Figure 7 at SWE = 0.5); for all HRUs with SDCs ($n = 54,104$) normalized SCA values less than 45% were classified as low ($n = 1,285$), greater than 55% were classified as high ($n = 51,334$), and between 45 and 55% were classified as mid ($n = 1,485$). These groups relate to the

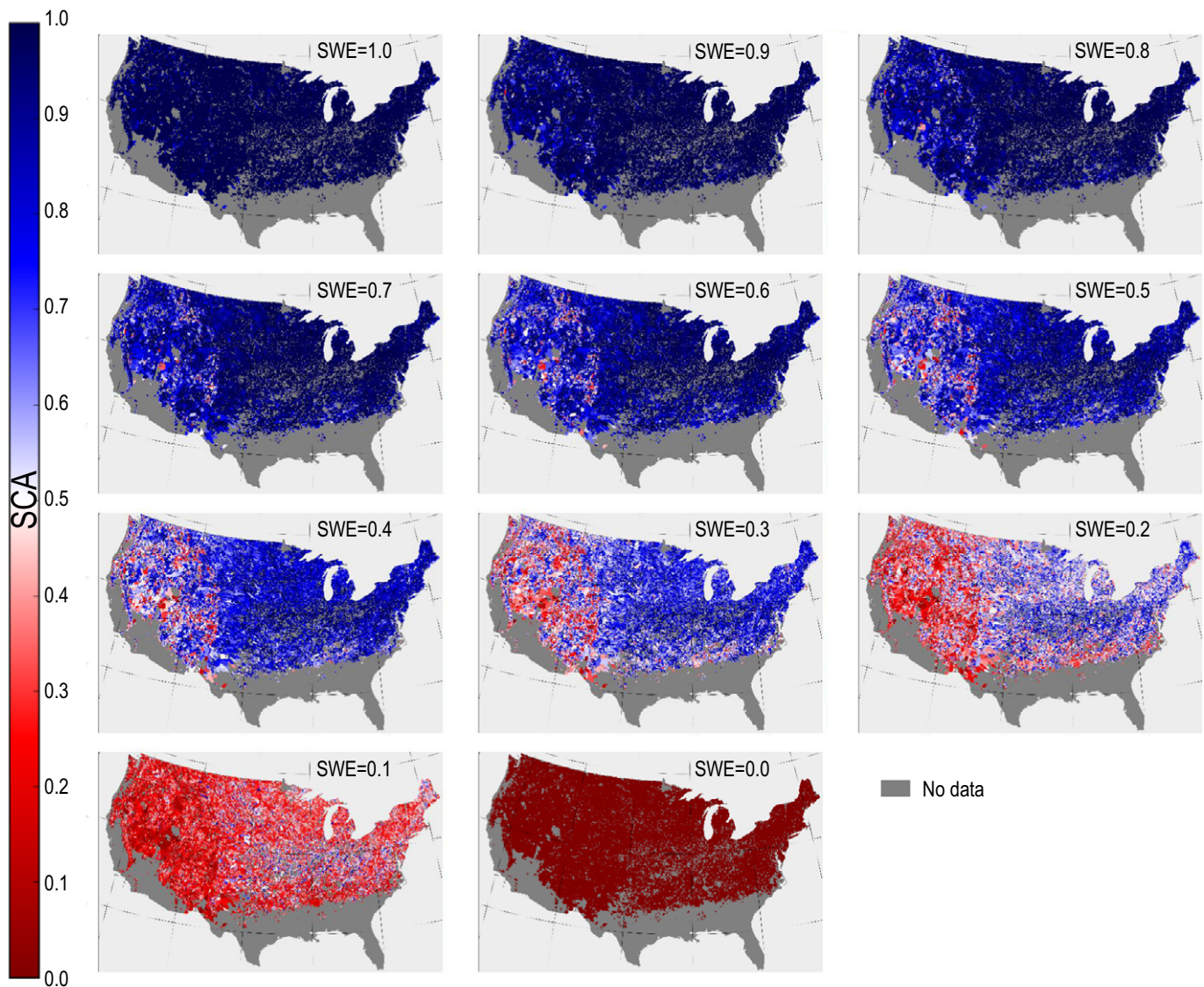


FIGURE 7. Progression of Calculated SCA for the Normalized SWE = 1 to SWE = 0 in 10% Increments for the Representative SDC for Each Study Watershed in CONUS.

colors in Figure 8: low SCA is shown in blue, high SCA is shown in red, and the mid-SCA is white. The SCA value at SWE = 0.5 was chosen to capture the variability in the SDC shape (e.g., Curve A or B from Figure 1); high SCA would be conceptually related to Curve A, while low SCA would be conceptually related to Curve B.

Relation of Topographic and Climatic Data to SDC Type

The distribution of HRU-scale topographic variables (elevation and slope) within these groups compared in boxplots (Figure 8). These data show that low and mid-SCA HRUs have substantially greater slopes than the

high SCA HRUs, and are also located at higher elevations, e.g., in the Mountain West. However, the elevations of low and mid-SCA groups were not substantially different than the high SCA group. The low and mid-SCA groups have similar distributions of elevation and slope with no substantial distinction between them. Aspect was also compared within these groups, but there was no substantial difference between groups.

The distribution of HRU-scale climatic variables (minimum and maximum temperature variance for early snowmelt months of March, April, and May) within these groups were compared in boxplots (Figure 9). Average monthly climate data (Thorton *et al.*, 2016) were also compared to evaluate relations between minimum and maximum temperature variability and the relation to SCA groups.

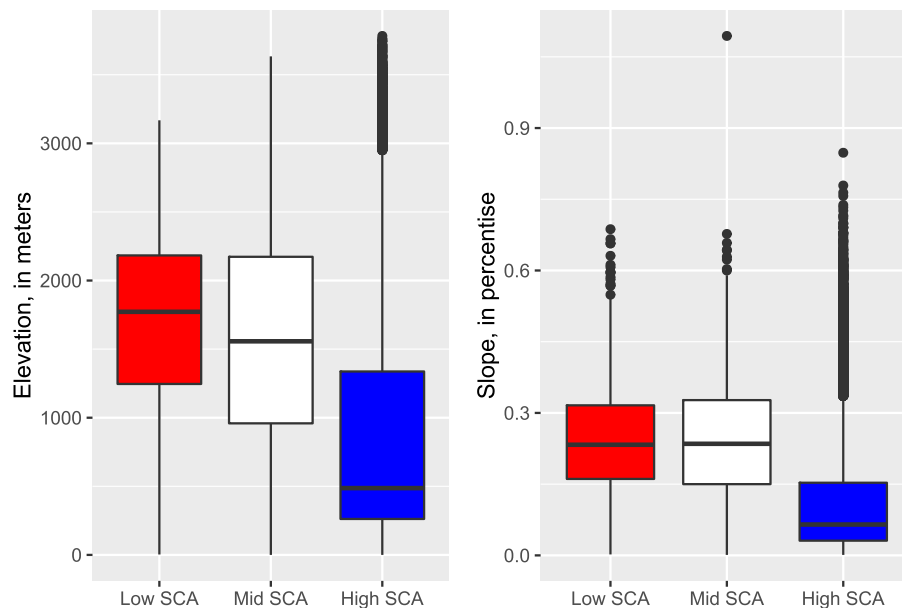


FIGURE 8. Distribution of Topographic Data (elevation and slope) for Each SDC Group.

Data show the variance of both the minimum and maximum temperature in March and April are substantially greater in the high SCA group relative to the low and mid-SCA groups, which again show very similar distributions. There is much less variability between the SCA groups for the May data.

DISCUSSION

Often snow processes are studied and described in the western U.S., where resources are more dependent on storage and distribution of runoff originating from snowmelt headwaters (Knowles *et al.*, 2006; Clow, 2010; Clow *et al.*, 2012; Jepsen *et al.*, 2012; Kalra *et al.*, 2013; Hedrick *et al.*, 2015; Broxton *et al.*, 2016); however, continental- and national-scale models require an analysis of SDCs for their entire spatial extents. The SNODAS provides a methodologically consistent spatial distribution of SCA and SWE data at the CONUS extent. SDCs were developed, using a consistent methodology to assess snowmelt processes on a national scale.

Improvement of representation of snow processes within broad-scale hydrologic models allow for assessment of water resources on a national scale (Kiang *et al.*, 2013; Archfield *et al.*, 2015). Process-based hydrologic models, such as the PRMS in the USGS's NHM, use watershed-scale HRUs to quantify hydrologic processes across the CONUS. Within PRMS, snow processes (snowmelt) are applied to the HRU-

scale using SDCs through the Snow Dynamics module (Markstrom *et al.*, 2015). In the past, PRMS used two default SDCs applied in the NHM: above treeline and below treeline. SDCs calculated for each HRU will improve NHM performance at the CONUS and local extents. Previous studies have challenged the assumption that there is a static annual SDC which represents snowmelt processes, however, these studies were conducted on significantly smaller scales (Luce and Tarboton, 2004; Dressler *et al.*, 2006; Shamir and Georgakakos, 2007; Jain *et al.*, 2010). This study developed and applied methods to create comparable SDCs at the HRU scale for all of the CONUS. The sensitivity to changes in SDCs within the NHM has not yet been quantified; however, this analysis will show where water resources might be most vulnerable to changes in snowmelt.

The relation of SDC groups to topographic and climatic variables show that broad categorization of SDCs could be extended beyond the spatial extent of SNODAS. The variance of the elevation, slope, and temperature the HRUs within the High SCA category may allow for classification of snowmelt processes independent of SNODAS or snow data. Representation of snowmelt processes in watershed-scale models in regions such as Alaska, which does not have SNODAS coverage, may be improved through these relations.

The challenge of representing the variability in snowmelt processes in watershed models through SDCs relies on the accuracy and relation of the SDCs to snowmelt processes. Climate change is driving earlier onset of snowmelt seasons (Clow, 2010), which has been shown to lengthen the overall snowmelt season (Barnhart *et al.*, 2016). As an integration of spatial

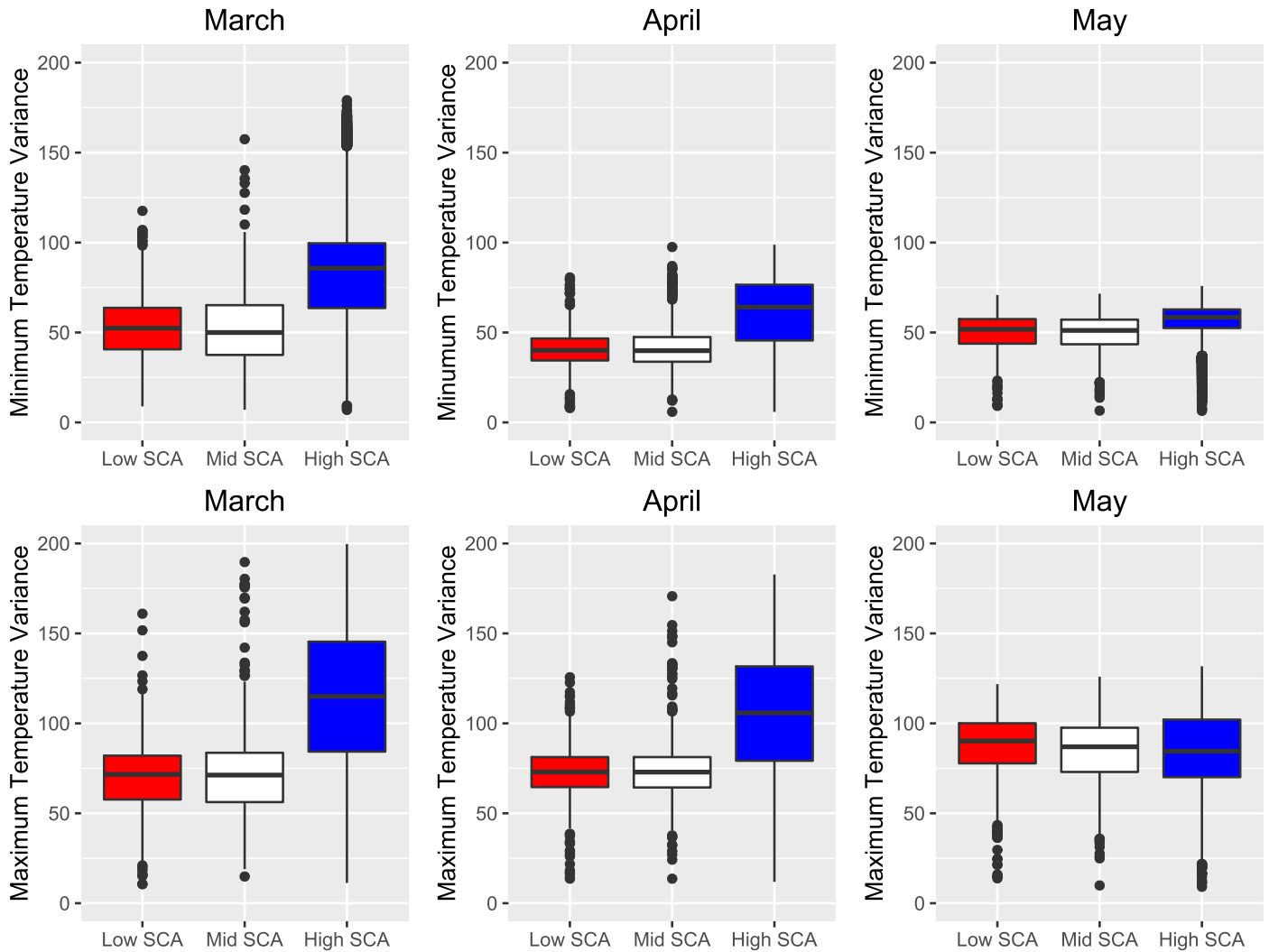


FIGURE 9. Distribution of Climate Variables (minimum and maximum temperature variance) for March, April, and May (left to right).

and temporal snowmelt processes, we expect SDCs will be affected by climate change; however, not all SDCs will be impacted in the same way. Areas where SWE declines at a faster rate than SCA (*e.g.*, Curve A) may not be as altered as areas where SWE declines at the same or a slower rate than SCA (*e.g.*, Curve B). Ecosystems and water users that rely on streamflow which originates from headwaters with SDCs which follow along Curve A could be more vulnerable to changes in the timing and duration of snowmelt.

CONCLUSIONS

Broad-scale hydrologic models are critical tools needed to inform international, national, and large

river basin water allocation, to achieve global water security, for national water assessments, to provide a consistent approach to evaluate water resources, to provide a foundation for international flood policy and water quality and ecological directives, and to plan for the effects of climate extremes on water resources (Archfield *et al.*, 2015). The evaluation of the spatiotemporal variability in SDCs is essential to assessment of water availability on a national scale. Relation of SDC groups to topographic and climatic variables allow for improved representation of snowmelt processes in physically-based hydrologic models in data-scarce regions. Quantification and categorization of snowmelt processes on a broad scale allows for identification of areas of possible increased vulnerability to climate change, which may be ideal candidates for more detailed, watershed-scale assessment.

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